

1. 解 由容斥原理: $P(A \cup B) = P(A) + P(B) - P(AB)$

由非负性: $P(A \cup B) \leq 1$

↑ 故 $P(AB) \geq 0.5$

↓ 即当 $A \cup B = \Omega$ 时有 $P(AB)$ 最小为 0.5

由非负性 $A \subset (A \cup B), B \subset (A \cup B)$

由单调性: $P(A \cup B) \geq P(A) = 0.7$

$P(A \cup B) \geq P(B) = 0.8$, 故 $P(A \cup B) \geq 0.8$

故 $P(AB) \leq 0.7$

即 $P(A \cup B) = 0.8$, 即 $A \subset B$ 时, 有 $P(AB)$ 最大为 0.7

解: 由容斥原理:

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(AB) - P(BC) - P(AC) + P(ABC) \\ &= \frac{1}{3} + \frac{1}{3} + \frac{1}{3} - \frac{1}{8} - \frac{1}{8} - 0 + 0 \\ &= \frac{3}{4} \text{ 即为所求} \end{aligned}$$

2. 解: 三局两胜制: 甲两局取胜: $P_1 = p^2$

三局取胜: $P_2 = C_2^1 p(1-p) \cdot p = 2p^2(1-p)$

故 $P_{\text{甲}} = p^2 + 2p^2(1-p) = (3-2p)p^2$

五局三胜制: 甲三局取胜: $P_1' = p^3$

四局取胜: $P_2' = C_3^2 p^2(1-p) \cdot p = 3p^3(1-p)$

五局取胜: $P_3' = C_4^2 p^2(1-p)^2 \cdot p = 6p^3(1-p)^2$

故 $P_{\text{甲}}' = p^3 + 3p^3(1-p) + 6p^3(1-p)^2 = p^3(6p^2 - 15p + 10)$

故 $P_{\text{甲}}' - P_{\text{甲}} = 6p^5 - 15p^4 + 12p^3 - 3p^2 = 3p^2(2p^3 - 5p^2 + 4p - 1)$

令 $f(p) = 2p^3 - 5p^2 + 4p - 1$, $f'(p) = 6p^2 - 10p + 4 = 2(3p - 2)(p - 1)$

由于 $p \in [0, 1]$ 且 $p \geq 0$

设当 $\frac{1}{3} < p < \frac{2}{3}$ 时, $f'(p) > 0$, $f(p) \uparrow$

$\frac{2}{3} < p \leq 1$ 时, $f'(p) < 0$, $f(p) \downarrow$

而 $p \in (\frac{1}{3}, 1]$ 时, $f(p) > f(\frac{1}{3}) = 0$

$$f(p) \geq f(1) = 0$$

故 $P_{甲} - P_{乙} > 0$, 乙的立场更不利

17. 解: 设公交车于 3:15, 3:30, 3:45, 4:00 准时到达

设 $A_1 = \text{"甲在 3:00 ~ 3:15 到"}$, $A_2 = \text{"甲在 3:15 ~ 3:30 到"}$,

$A_3 = \text{"甲在 3:30 ~ 3:45 到"}$, $A_4 = \text{"甲在 3:45 ~ 4:00 到"}$.

B_1, B_2, B_3, B_4 对乙同理

$$\begin{aligned} \text{两人同乘一列车的概率 } P &= P(A_1 B_1) + P(A_1 B_2) + P(A_2 B_3) + P(A_3 B_4) \\ &= \frac{1}{4} \times \frac{1}{4} + \frac{1}{4} \times \frac{1}{4} + \frac{1}{4} \times \frac{1}{4} + \frac{1}{4} \times \frac{1}{4} \\ &= \frac{1}{4} \end{aligned}$$

然而本题未约定公交车会于 3:00 发一列车, 故若并非整点钟发车, 则问题结果将无法确定, 故约定为上述情况假设情况, 结果为 $\frac{1}{4}$

21. 解: $\Delta = B^2 - 4C$

有实根需要 $\Delta \geq 0$, 满足的 (B, C) 有:

$\{6, 1\}, \{6, 2\}, \{6, 3\}, \{6, 4\}, \{6, 5\}, \{6, 6\}, \{5, 1\}, \{5, 2\}, \{5, 3\},$

$\{5, 4\}, \{5, 5\}, \{5, 6\}, \{4, 1\}, \{4, 2\}, \{4, 3\}, \{4, 4\}, \{3, 1\}, \{3, 2\},$

$\{2, 1\}$

$$\text{故 } P(\text{方程有实根}) = \frac{19}{6 \times 6} = \frac{19}{36}$$

有重根需要 $\Delta = 0$, 满足的 (B, C) 有: $\{4, 4\}, \{2, 1\}$

则 $P(\text{方程有重根}) = \frac{2}{6 \times 6} = \frac{1}{18}$

解: (1) ~~第1次~~ 设 $A = \text{"取到第一箱"}$, $B = \text{"第一次取到一等品"}$

则 $P(B|A) = \frac{1}{5}$, $P(B|\bar{A}) = \frac{3}{5}$, $P(A) = \frac{1}{2}$, $P(\bar{A}) = \frac{1}{2}$

则 $P(B) = P(A)P(B|A) + P(\bar{A})P(B|\bar{A}) = \frac{1}{2} \times \frac{1}{5} + \frac{1}{2} \times \frac{3}{5} = \frac{2}{5}$

(2) 设 $C = \text{"第二次取到一等品"}$

则 $P(BC|A) = \frac{10 \times 9}{50 \times 49} = \frac{9}{245}$, $P(BC|\bar{A}) = \frac{18 \times 17}{30 \times 29} = \frac{51}{145}$

则 $P(BC) = P(A)P(BC|A) + P(\bar{A})P(BC|\bar{A}) = \frac{1}{2} \times \frac{9}{245} + \frac{1}{2} \times \frac{51}{145} = \frac{276}{1421}$

则 $P(C|B) = \frac{P(BC)}{P(B)} = \frac{690}{1421}$

证: A 倾向于 B , 则 $P(B|A) > P(B)$

即 $\frac{P(AB)}{P(A)} > P(B)$, 即 $P(AB) > P(A)P(B)$

证: $P(\bar{A}\bar{B})$ ~~证~~ $\frac{P(\bar{A}\bar{B})}{P(\bar{A})} > P(\bar{B})$ De Morgan $P(\overline{A \cup B})$

$= 1 - P(A \cup B) \stackrel{\text{容斥}}{=} 1 - [P(A) + P(B) - P(AB)]$

$= 1 - P(A) - P(B) + P(AB) > P(A)P(B) - P(A) - P(B) + 1$

$= (1 - P(A))(1 - P(B)) = P(\bar{A})P(\bar{B})$

即 $\frac{P(\bar{A}\bar{B})}{P(\bar{A})} > P(\bar{B})$, 即 $P(\bar{B}|\bar{A}) > P(\bar{B})$, 即 $\bar{A}$ 倾向于 $\bar{B}$. $\square$